

Cover bins sketch

Coverage closes when defined bins get hits

Sample, hit, hole

- You sample a value into the coverpoint; matching bins increment
- Coverage percent is bins hit over bins defined
- Hitting mid once on a four-bin nibble point is twenty-five percent with three holes left

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Coverpoint / bins sketch

Document-aid literacy: a **coverpoint** partitions values into **bins**; sampling hits bins and leaves **holes**. Not a simulator coverage DB. Starter: nibble coverpoint with mid already hit.

Starter example: nibble coverpoint with bins low/mid/high/top; sample 5 already hit mid. Sample more values to close holes.

Load starter example

Challenges 0 / 22 cleared

Quiz: **coverpoint**: A coverpoint usually measures...

☐ how a field/expression's values were observed (via bins)

☐ FPGA place-and-route congestion

☐ SDF interconnect delay only

☐ the Makefile target name

Show hint

Check

Next

Idle

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Core ideas

Coverpoint

A field (or expression) you care about.

Bins

Named buckets over that field's values.

Sample

A observed value hits one bin.

Hole

A defined bin never hit yet.

Lab

Lab starter

Planning docs practice

- Sketch one coverpoint with four named bins for a data nibble or opcode set
- Mark which bin you would hit with a directed sample first
- Note one value that sits outside your bins and decide whether to add a bin or ignore it

Pitfalls to watch

- Do not confuse “more random” with hole closure
- Do not redefine holes as every unseen number in nature, only defined bins count
- Do not skip naming bins because cross coverage feels advanced; start with clear singles
- And remember the lab is a sketch, not your simulator’s coverage DB

Your turn

- Complete the checklist for at least one track, preferably both
- Close or name holes on a small bin sketch